Replicating Finkelstein (2002) on bonds, equity options and cash equity

A build plan

Prepared 30 October 2026. A method plan. Nothing here has been backtested.

Read this first

Next action. Build work package WP1 in Section 10. It needs no market data, and its expected numbers are already in this document.

The answer.

1. The deck gives a model and a hedge book. It gives no trading rule, so every trading rule below is an addition.
2. The model is understood. I rebuilt it from the slides. At an assumed rate of 5%, it reproduces the GX skew table within 0.14 vol points, the GX credit curve within 17 bp of my chart readings, and all 12 hedge numbers within 7.2%.
3. Slide 1 does not reproduce as printed. The two put prices fit a rate near 4.8%, not the printed 3%. The 1.42 does not follow the formula that gives 1.90. Section 9 has the list.
4. The deck already uses a bond as its market credit curve (slide 14). So the bond leg extends the deck's own test. It is not a departure from it.
5. Three markets give two independent gaps. That identity decides which trade to do, and it is the main addition in this plan.
6. A 2026 *Journal of Finance* paper now tests the bond-against-stock gap directly. High-gap bonds outperform and high-gap stocks underperform, which is the direction of trade T1. It ranks names each month, uses realised vol, and reports gross returns. So I changed the base signal from a fixed trigger to a monthly rank, and made its replication the first empirical run.

Conventions. These apply unless a test uses the deck's own values.

- "Slide N" is the number printed on the slide. PDF page = slide + 1. The brief's "Page 1" is slide 1.
- Spreads are in basis points. Probabilities are decimals. Duration is in years. Prices are per 100 of face, the deck's scale.
- Recovery on senior unsecured bonds is 0.40. The deck itself uses 0.30 and 0.10 on GX, and Yu uses 0.50.
- Coupons are semiannual. Day count is 30/360 on corporates, ACT/ACT on Treasuries, ACT/360 on repo and stock loan.
- Zero rates and Z-spreads are continuously compounded. Option time is ACT/365.

Provenance tags. [DECK s.N] is the deck, slide N. [CREDITGRADES], [LARDY], [YU] and [OWN FILES] name other sources. [ADAPTATION] is my own reasoning, and each one gives a reason. No adaptation is Finkelstein's result. Check results come from `check_deck_numbers.py`, with output in `check_deck_numbers.out`. The page-by-page extraction is `01_deck_extraction.md`.

1 What the method computes

Finkelstein treats the stock as a shifted lognormal variable with a knock-out: $S_t = (V_t - B)\, e^{\int_0^t (r-p)}$. The asset value V has zero drift, and default happens when V first touches the barrier B (slides 4, 5). Equity options set the term structure of asset volatility. The skew sets B when low-strike puts are liquid. Otherwise $B = \bar{L}D$, with $\bar{L}$ near 0.5 (slides 9, 11). The output is a closed-form survival curve $Q(T)$ at every tenor (slides 11, 18). An assumed recovery turns that curve into CDS par spreads and bond prices (slide 12). The same model prices every option, so a put splits into a default part and a diffusion part (slides 1, 6). The deck tests this on one name, GX, against a stock skew, a bond and a bond-generated CDS curve (slides 8, 12, 14, 15). It then solves hedges in stock, puts and calls against delta, vega and jump to default (slides 21–25). **The deck gives no arbitrage signal and no trading rule.** It has no entry, exit, sizing, cost or backtest. Its only related words are "trading framework" (slide 2), "relative value analysis" and "the residual noise between the formula's credit

spread and the market is diversifiable like a white noise" (slide 26). So the model-against-market gap is the signal the deck implies. Every rule built on it here is tagged as an addition.

2 How it differs from E2C, CreditGrades and Yu

The biggest difference: Finkelstein takes vol and the barrier from equity options, and one model prices both credit and options. E2C, CreditGrades and Yu use a single flat asset vol and a fixed or spread-fitted barrier. None of them can price a put. All four share one survival formula. Slide 3 and Lardy's printed slide 36 name the same four authors for it, and it matches `e2c_bond.survival` exactly at constant vol. Finkelstein and Lardy are also named authors of the CreditGrades document, so these are versions of one idea, not independent checks of it. CreditGrades' bibliography also lists an earlier talk with this deck's title (Finkelstein, 2001), one with Lardy (2001), and a JPMorgan patent filing document (Lardy, Finkelstein, Khuong-Huu and Yang, 2000). I did not find any of them in your files. They may print what this deck leaves out, such as the closed-form option formula.

	Finkelstein 2002	**Lardy E2C 2002**	**CreditGrades 2002**	**Yu 2005**
Asset vol	Term structure $\sigma_{\text{cum}}(T)$, bootstrapped from options (s.11, 12)	Flat. From reference S^*, σ_s^* (printed 7). Flat in-sample vol per name in his tests (printed 16)	Flat. 1,000-day historical (p.18)	Flat. 1,000-day historical (PDF p.17)
Barrier	Skew if liquid, else $\bar{L}D$, $\bar{L} \approx 0.5$. GX is $0.42D$ (s.8, 9)	$\bar{L} = 0.5$ from a recovery database (printed 6)	$\bar{L} = 0.5$	$\bar{L}$ fitted on first 10 CDS days (PDF p.18)
λ	Charts at 0 to 0.4, none chosen (s.19). My fit: 0 on GX, near 0.28 on the hedge example	0.3	0.3	0.3
Short end	Flat intensity below 0.5y, so $Q(0) = 1$ (s.19)	Keeps the mass $1 - B(0)$ (printed 7)	Keeps it (A.6)	Keeps it (Eq. 12)
Options	Priced as knock-outs. Used as inputs and hedges (s.6, 8, 21–25)	Named as a use, no formula (printed 38)	Not priced. Case study lists puts as an alternative (pp.35–36)	Not used. Implied vol named as untried (PDF p.11)
Credit input	CDS. On GX, a CDS curve built from a bond (s.14)	CDS	CDS, one bond spread case (p.33)	5-year CDS
Evidence	Two names: GX, EXDS (s.8–15)	142 names, R^2 89% (printed 16–17)	122 firms, 6,194 quotes (Ch. 3)	261 obligors, 135,759 spreads (PDF p.15)
Hedge	Delta, vega, jump solved in stock, puts, calls (s.21–25)	Equity equivalents (printed 32–33)	Delta, gamma, vega curves. WorldCom 7.7:1 (pp.14, 36)	Delta from Eq. 15, fixed at entry
Trading rule	**None**	Named "arbitrage", no rule (printed 38)	Case studies, no rule	**Yes.** Eq. 8, exits, capital, costs

Your own workbooks differ again. They trade the residual of a weekly cross-section of log implied-over-historical asset vol on $\ln(S/D)$ (`notes/own_models/02_implemented_method.md`). Yu asks if a name is cheap against its own history. The workbooks ask if it is cheap against peers at the same leverage. Finkelstein's model can feed either one.

What later papers change

I read each of these in full text. Numbers below come from the pages cited. None of them tests Finkelstein's model itself.

Paper and copy read	What it finds	What changes in this plan
Chen, Chen and Li (2026). Published version, open access	The gap between actual and equity-implied bond spreads (DES) predicts bonds up and stocks down. Model is CreditGrades extended with payout. Vol is 3-year realised. No options (pp.2006–2007, 2015)	Base signal becomes a monthly rank (step 24a, trade T0). Replicate this first. Exit by time, not convergence
Avino and Salvador (2024). Accepted manuscript of 5 January 2024, so published numbers may differ	Put-only hedge of market value, not of default loss. Slope on the model ratio is 1.02. Hedged error falls 19% with puts and 14% with stock (pp.10, 23, 29)	Add the put-only size to steps 26, 27 and T2. Separate options used to calibrate from options used to hedge
Cao, Goyal, Xiao and Zhan (2023). Full text in your folder	Bonds with the largest rise in implied vol lag by 0.60% a month. That rise does not predict stock returns. Net of cost only for large trades (pp.1375–1377, 1394)	Add step 24b as a filter on long-bond trades, with no stock leg
Bégin, Boudreault and Thériault (2023). Full text in your folder	Put-implied intensity plus CDS gives recovery: $R = 1 - \text{CDS}/\lambda$. Put-implied default probability is more volatile than CDS-implied (pp.125, 127)	Add a flat-hazard form of step 20, and expect the option side to be noisier

3 The method as a pipeline

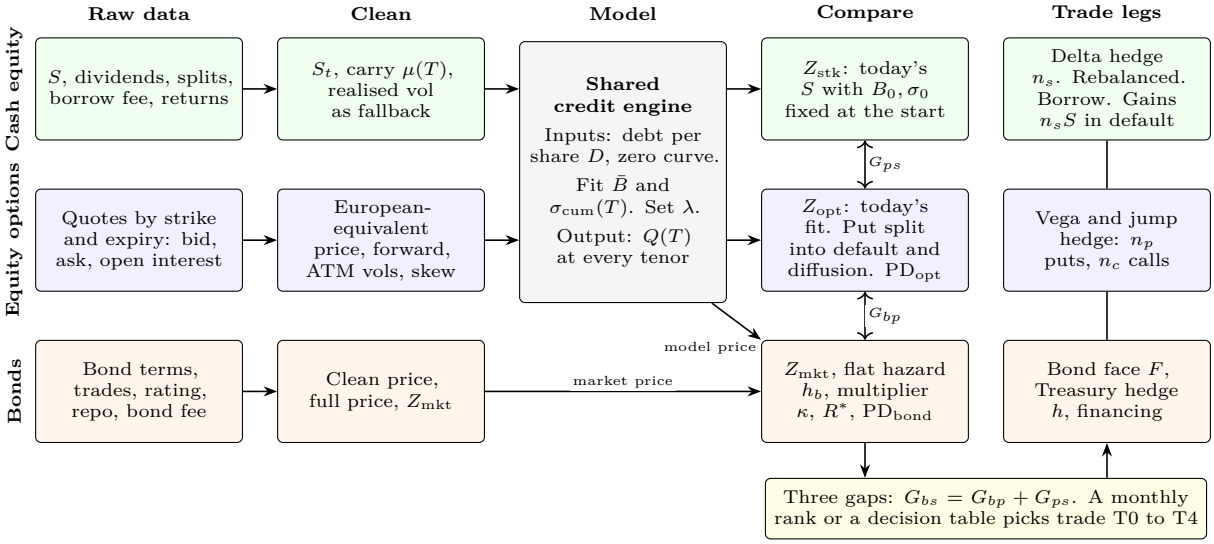


Figure 1: How the three instruments connect. Read left to right. Options set the credit engine. The bond's price tests it. Stock and options hedge the result.

1. **Cash equity supplies S, carry and the delta hedge.** S fixes the distance to the barrier through $V_0 = S_0 + B$. Realised vol is only a fallback here. The deck's vol is implied (slide 26).

2. **Equity options supply vol, skew and an implied default probability.** ATM vols by maturity bootstrap $\sigma_{\text{cum}}(T)$. The skew identifies B. A put contains a default part: the deck's 1.90 of 4.92 at the money, and 1.42 of 2.72 at 30% out of the money.

3. **The credit engine turns those into $Q(T)$.** Q prices any cash flow, so maturity mismatch between options and bonds disappears.

4. **Bonds supply the market credit spread, in place of CDS.** The deck does this itself on slide 14. The same Q prices the bond's own cash flows and gives a model Z-spread to set against the market Z-spread.

5. **The gaps say which market is out of line.** With stock and fixed parameters on one side, and today's options on another, two gaps are independent (Section 8).

6. **Greeks and jump to default come from the same model.** They size the stock, put and call legs (slides 20–25).

7. **Trades, P&L, exits and reports are additions.** The deck has none of them.

4 Master table of raw inputs

All names are neutral. "Source" is the instrument or data family, not a vendor. E, O, B, F and R label the groups used in Section 5.

Name	Meaning	Unit	Type	Frequency	Source
E1 `stock_close`	Closing share price	USD/share	float	daily	Cash equity
E2 `split_factor`	Split and stock dividend adjustment	ratio	float	event	Cash equity
E3 `dividend`	Cash dividend, by ex-date	USD/share	float	event	Cash equity
E4 `shares_out`	Common shares outstanding	count	int	quarterly, event	Cash equity
E5 `borrow_fee`	Stock loan fee	decimal/yr	float	daily	Cash equity
E6 `hard_to_borrow`	Name flagged hard to borrow	flag	bool	daily	Cash equity
E7 `stock_half_spread`	Half bid-ask on the stock	fraction	float	daily	Cash equity
E8 `equity_id`	Issuer link to options and bonds	id	string	static	Cash equity
O1 `option_id`	Underlying, expiry, strike, call or put	id	string	static	Options
O2 `bid`, `ask`	Closing option quotes	USD/share	float	daily	Options
O3 `quote_time`	Time of the quote, to align with S	time	time	daily	Options
O4 `volume`, `open_interest`	Liquidity of the contract	contracts	int	daily	Options
O5 `exercise_style`	American or European	enum	enum	static	Options
O6 `multiplier`	Contract size and any adjustment	shares	int	static	Options
O7 `vendor_iv`, `vendor_delta`	Optional cross-check only	decimal	float	daily	Options
B1 `bond_id`, `issuer_id`	Identifier and issuer link	id	string	static	Bonds
B2 `coupon`, `freq`	Coupon rate and payments a year	decimal, int	float, int	static	Bonds
B3 `maturity`, `issue_date`	Dates	date	date	static	Bonds
B4 `day_count`, `first_coupon`	Accrual conventions	enum, date	mixed	static	Bonds
B5 `seniority`	Senior unsecured or other	enum	enum	static	Bonds
B6 `call_schedule`, flags	Call dates and prices. Convertible, putable, sinking, floating flags	USD, flag	mixed	static	Bonds
B7 `amount_out`	Face outstanding	USD mm	float	event	Bonds
B8 `trade_price`, `trade_size`	Each customer trade report	% of par, USD	float	intraday	Bonds
B9 `trade_time`, `report_time`	Execution and publication time	time	time	intraday	Bonds
B10 `side`, correction fields	Buy or sell. Reversal and correction links	enum	mixed	intraday	Bonds
B11 `rating`	Rating known on the date	enum	enum	event	Bonds
B12 `repo_rate`, `bond_fee`	Financing of a long. Borrow cost of a short	decimal/yr	float	daily	Bonds
F1 `st_debt`, `lt_debt`	Short-term and long-term borrowings	USD mm	float	quarterly	Accounts
F2 `other_st`, `other_lt`	Other short-term and long-term liabilities	USD mm	float	quarterly	Accounts
F3 `minority`, `preferred`	Minority interest, preferred equity	USD mm	float	quarterly	Accounts
F4 `total_liabilities`	For Yu's alternative D	USD mm	float	quarterly	Accounts
F5 `filing_time`	When each filing became public	time	time	quarterly	Accounts
R1 `par_yields`	Treasury par yield by tenor	decimal	float	daily	Rates
R2 `cash_rate`	Short rate for stock and option financing	decimal/yr	float	daily	Rates
R3 `treasury_price`	Price of the Treasury used as hedge	USD/100	float	daily	Rates
V1 `events`	Merger, tender, call, default dates	date	date	event	Events
V2 `realised_recovery`	Price after default, where known	% of par	float	event	Events
V3 `cds_par`	Optional. CDS par spread by tenor, as a check only	bp	float	daily	Credit, optional

Assumptions, not data. $\bar{L}$ (0.5 [DECK s.9]), λ (0.3 [LARDY] [CREDITGRADES]), R_b (0.40), triggers, and the cost

numbers. Section 11 ranks the choices.

5 Every step

New code goes in `fk_*.py` modules. `e2c_bond.py` is imported, never edited. Its `survival`, `risky_bond_price`, `default_leg` and `run_bond_backtest` do not fit, because they assume one flat vol through `Firm`. Its helpers below do fit. They price per 1 of face, and this plan's tables are per 100.

Stage A. Clean the market data

Step 1. `equity_panel` [ADAPTATION]

Purpose Build the daily stock series.
Inputs E1–E4, E8.
Formula Adjust S for splits only. $x_t = \ln\left((S_t + \text{div}_t)/S_{t-1}\right)$.
Outputs S_t, div_t, x_t, shares. Used by 5, 7, 11, 12, 26, 32.
Why The deck takes S as given. Corporate actions follow Yu's footnote 14 through `debt_after_corporate_action`.

Step 2. `zero_curve_from_par_yields` [ADAPTATION]

Purpose Build the discount curve. Reused from `e2c_bond`.
Inputs R1.
Formula Bootstrap continuously compounded zeros.
Outputs `ZeroCurve`, $\text{DF}(t)$. Used by 4, 5, 14, 15, 16, 17.
Why The deck uses one flat rate. A bond needs each cash flow discounted at its own date. $Q(T)$ does not depend on the curve, so a full curve is allowed.

Step 3. `clean_option_quotes` [ADAPTATION]

Purpose Remove quotes that cannot be priced.
Inputs O1–O6, S_t.
Formula Keep bid > 0, ask $>$ bid, open interest > 0, (ask $-$ bid)/mid ≤ 0.5, expiry ≥ 30 days. Drop any quote that breaks: put price rises with K, convex in K, $P(K)/K$ rises with K.
Outputs Clean chain. Used by 4, 5, 6, 22.
Why Carr and Wu (2011, p.482) use bid > 0 and open interest > 0. The other bounds are my judgement.

Step 4. `european_equivalent` [ADAPTATION]

Purpose Turn an American quote into a European price and a Black-Scholes vol.
Inputs Clean mids, S_t, curve, dividends, borrow fee.
Formula Solve $\text{Tree}_{\text{Am}}(v) = \text{mid}$ with discrete dividends. Then $P_E = \text{BS}(v)$.
Outputs P_E, $\sigma_{BS}(K,T)$, Δ_{BS}. Used by 5, 6, 8, 10, 22.
Why Listed single-stock options are American (Carr and Wu, p.481). The deck prices with Black-Scholes. The tree has no default, so it may understate the early-exercise value. WP8 measures that.

Step 5. `implied_forward` [ADAPTATION]

Purpose Get the carry the option market actually uses.
Inputs P_E, C_E near the money, $\text{DF}(T)$.
Formula Regress $C_E - P_E = \text{DF}(T)(F - K)$ across strikes. $\mu(T) = \ln(F/S)/T$.
Outputs $F(T)$, $\mu(T)$, an implied borrow estimate $r - p_{\text{known}} - \mu$. Used by 8, 10, 16, 29.
Why Slide 4 calls r a stock financing rate and p a dividend yield. The forward bundles both, and it reveals a high borrow fee.

Step 6. `select_surface` [DECK s.8] [ADAPTATION]

Purpose Choose the points used for calibration.
Inputs $\sigma_{BS}(K,T)$, Δ_{BS}, spreads, open interest.
Formula Maturities 0.5 years and longer. Strikes with $K/F \in [0.4, 1.4]$. Out-of-the-money side only. Weight $w = 1/$ half-spread in vol. Need 3 or more strikes below the money in 2 or more maturities for a skew fit.
Outputs Table of (K, T, σ, w). Used by 8, 10.

Why The GX table spans K/S of 0.48 to 1.43 at 1.59 and 2.59 years. The thresholds are my judgement. The deck has no rule.

Stage B. Calibrate the firm

Step 7. `debt_per_share` [YU] [OWN FILES]

Purpose Compute D. Reused from `e2c_bond` for the default form.
Inputs F1–F5, E1, E4.
Formula Default: D = total liabilities / shares. Variant: FinD = ST + LT + 0.5(OtherST + OtherLT), $k_0 = \min(\text{MI}, 0.5\,\text{FinD})$, $D = (\text{FinD} - k_0)/\big((\text{MktCap} + \text{Pref})/S\big)$. Use `filing_usable_from` for timing.
Outputs D_t, both forms. Used by 9, 11, 13a, 25.
Why The deck says "current debt per share" with no formula (s.8, 9). Yu and Chen, Chen and Li (p.2015) both use total liabilities, so start there. The variant matches your master file on all 19,489 rows. D matters more here than in Yu, because no spread fit absorbs its level.

Step 8. `fit_barrier_to_skew` [DECK s.8, 9, 11] [ADAPTATION]

Purpose Read the barrier from the skew, using one B across maturities.
Inputs Step 6 table, $\mu(T)$, DF(T).
Formula $\min_{B,\sigma_1..\sigma_m} \sum_{i,j} w_{ij}\big(\sigma^M(K_i, T_j; B, \sigma_j) - \sigma_{ij}\big)^2$, with σ^M from step 16.
Outputs B_{skew}, σ_j, rmse, status. Used by 9.
Why The deck uses "the same default barrier level" for two maturities and prints no objective. On GX market vols this returns $B = 6.01$, against the deck's 6.

Step 9. `choose_barrier` [DECK s.9, 11] [ADAPTATION]

Purpose Pick the barrier to use.
Inputs B_{skew}, rmse, D, $\bar{L} = 0.5$.
Formula $\bar{B} = \bar{L}D$ by default. Use B_{skew} only if it cuts rmse by 0.5 vol points or more, and $0.25D \le B_{\text{skew}} \le D$. Hold $\bar{B}$ fixed until the next filing.
Outputs $\bar{B}_t$, source flag. Used by 10, 12, 15, 16, 23.
Why The deck allows either route. B is weakly identified: on GX, rmse stays within 0.23 vol points for B from 5 to 7. The 2.59-year default probability moves by 0.026 over that range, so the error does little harm.

Step 10. `bootstrap_asset_vol` [DECK s.11, 12] [ADAPTATION]

Purpose Build the asset vol term structure.
Inputs ATM $\sigma_{BS}(T_j)$, $\bar{B}$, $\mu(T)$.
Formula Solve $\sigma^M(F, T_j; \bar{B}, \sigma_{\text{cum}}(T_j)) = \sigma_{BS}(F, T_j)$. Total variance $v_j = \sigma^2_{\text{cum}}(T_j)T_j$. Forward variance $(v_j - v_{j-1})/(T_j - T_{j-1})$, which must be positive. Linear in total variance between maturities. Flat forward past the last one.
Outputs $\sigma_{\text{cum}}(T)$ and the forward vols. Used by 12, 13, 16.
Why Slide 12 says "boot-strapped" and stops there. Solving the model beats the slide 6 rule: at GX the rule gives 52.3%, against the deck's 49.8%. The GX forward is 40.7%. With it the slide 12 curve reproduces within 17 bp.

Step 11. `asset_vol_fallback` [OWN FILES] [CREDITGRADES]

Purpose Give a vol when a name has no usable options.
Inputs x_t, S, $\bar{B}$.
Formula $\sigma = \sigma_S\, S/(S+\bar{B})$. σ_S is the mean of 30, 60, 90, 180 and 360 day realised vols. Reuse `historical_equity_vol` per window. Variant for step 13a: 3-year daily σ_S, then $\sigma = \sigma_S(1 - R_0)f$ with $R_0 = D/(S + D)$ and $f = 1$, 1.05, 1.10, 1.20, 1.40, 1.80 at breaks 0.25, 0.35, 0.45, 0.55, 0.75.
Outputs Flat σ_{cum}, flagged as fallback. Used by 12.
Why The deck has no fallback. In the rerun of your workbooks the realised blend gave Sharpe 2.49, and 162-week vol gave 0.15. CreditGrades and Yu use 1,000 days. The two own-file numbers are gross and in-sample. The variant is the rule in Chen, Chen and Li (p.2015). It gives $0.60\sigma_S$ against CreditGrades' $0.67\sigma_S$ at $S/D = 1$, and $0.36\sigma_S$ against $0.33\sigma_S$ at $S/D = 0.25$.

Stage C. Survival curve

Step 12. `fk_survival` [DECK s.11, 18]

Purpose Survival probability at one tenor.
Inputs T, S_0, $\bar{B}$, $\sigma_{\text{cum}}(T)$, λ.
Formula $Q(T) = N\big(\frac{\ln \bar{d}}{A_T} - \frac{A_T}{2}\big) - \bar{d}\, N\big(-\frac{\ln \bar{d}}{A_T} - \frac{A_T}{2}\big)$, with $A_T = \sqrt{\sigma^2_{\text{cum}}(T)T + \lambda^2}$ and $\bar{d} = \frac{S_0+\bar{B}}{\bar{B}} e^{\lambda^2}$. At $\lambda = 0$ this is slide 11.
Outputs $Q(T)$. Used by 13.

Step 13. `fk_curve` [DECK s.19]

Purpose Survival on a grid, with the short-end remedy.
Inputs Step 12 at $T \geq 0.5$.
Formula For $T < 0.5$ and $\lambda > 0$: $Q(T) = Q(0.5)^{T/0.5}$, which is flat intensity from 1. Grid of 1/48 year.
Outputs Q on the grid. Hazard $h(T)$. Used by 14, 15, 19–23.

Step 13a. `des_survival` [ADAPTATION]

Purpose Survival in the extended CreditGrades model, to replicate the published gap.
Inputs $V_0 = S_0 + \bar{L}D$, $\bar{L} = 0.5$, $\lambda = 0.3$, σ from step 11 variant, r, payout δ.
Formula $q(t) = \int_{-\infty}^{\bar{z}} q(t|z)\phi(z)\, dz$, $\bar{z} = \frac{\lambda}{2} - \frac{1}{\lambda}\ln\frac{\bar{L}D}{V_0}$. $q(t|z) = N\big(C\sigma\sqrt{t} - \frac{y}{\sigma\sqrt{t}}\big) - e^{2Cy} N\big(C\sigma\sqrt{t} + \frac{y}{\sigma\sqrt{t}}\big)$, with $C = \frac{r-\delta}{\sigma^2} - \frac{1}{2}$ and $y = -\ln\frac{V_0}{\bar{L}D} - \frac{\lambda^2}{2} + \lambda z$. δ = (dividends + interest + repurchases)/$(S + D)$, capped at 0.15.
Outputs $q(t)$. Used by 15, 23, 24a.
Why This is Chen, Chen and Li (Eqs. 1–9, pp.2012–2014). It differs from slide 18 in two ways: V drifts at $r - \delta$, and barriers above V_0 are cut off. At zero drift and $T = 1$ it gives 0.936004 against Eq. (1)'s 0.934494 at $S/\bar{B} = 1$. Keep the deck's zero-drift form for all option work.

Stage D. Model values

Step 14. `fk_cds` [DECK s.12] [LARDY] [CREDITGRADES]

Purpose CDS legs and par spread, for tests and for sizing.
Inputs Q, DF, R, premium frequency.
Formula $L = \sum_j \text{DF}(\text{mid}_j)(Q_{j-1} - Q_j)$. $A = \sum_i \Delta_i \text{DF}(t_i) Q(t_i) + \frac{1}{2}\Delta L$. Par $= (1 - R)L/A$. Optional $\times 360/365$ (CreditGrades, printed p.9).
Outputs A, L, par. Used by 26, 31.
Why The deck prints no CDS formula. Lardy's printed slide 7 gives the integral form. Deck charts fit it at $r = 5\%$.

Step 15. `fk_bond_price` [DECK s.12] [ADAPTATION]

Purpose Value this bond's own cash flows on the model curve.
Inputs Bond terms, Q, DF, R_b. `cash_flows` from `e2c_bond`.
Formula $P = \sum_i CF_i\, \text{DF}(t_i)\, Q(t_i) + R_b\, L(T)$, then Z from `z_spread`.
Outputs $P_{\text{opt}}, Z_{\text{opt}}$ and, with step 23's curve, $P_{\text{stk}}, Z_{\text{stk}}$. Used by 24, 26.
Why Slide 12 says it computes bond prices and prints no formula. This is the form of `risky_bond_price`. Recovery is of par, so a discount bond loses less than a premium bond. With flat 42% vol at $r = 5\%$ it gives clean prices 83.4, 59.9, 44.5 against the deck's 82, 60, 43.

Step 16. `fk_put` [DECK s.1, 6] [ADAPTATION]

Purpose Price a put and split it into default and diffusion parts.
Inputs $S_0, K, T, \sigma_{\text{cum}}(T), B, r, p$.
Formula At $\lambda = 0$: default part $= K\, e^{-rT}(1 - Q(T))$. Diffusion part $= e^{-pT}\big[G(V_0) - \frac{V_0}{B} G(B^2/V_0)\big]$, where $H = B + Ke^{-(r-p)T}$ and $G(x) = H[N(-d_2(x,H)) - N(-d_2(x,B))] - x[N(-d_1(x,H)) - N(-d_1(x,B))]$. $d_1(x,y) = \big(\ln(x/y) + \frac{1}{2}\sigma^2_{\text{cum}}T\big)/(\sigma_{\text{cum}}\sqrt{T})$, $d_2 = d_1 - \sigma_{\text{cum}}\sqrt{T}$. Call by parity. Invert for σ^M.
Outputs Put, default part, diffusion part, $\sigma^M(K,T)$. Used by 8, 10, 22, 26.
Why Slide 6 says closed form and prints none. This is the reflection principle. It reproduces slide 8 within 0.14 vol points and matches direct integration to 8 decimals.

Stage E. Market side and gaps

Step 17. `market_z` [ADAPTATION]

Purpose Get the market spread of the bond.
Inputs B1–B10, DF.
Formula Reuse `clean_trade_reports`, `fix_decimal_shifts`, `flag_price_spikes`, `daily_clean_price`, `full_price`, `z_spread`. Screen with `bond_terms_are_eligible`, `maturity_is_eligible`, `passes_liquidity_screen`.
Outputs Full price, Z_{mkt}. Used by 18–21, 24.
Why This is Part 3 of `notes/paper_method/02_bond_vol_method.md`. Z against Z compares like with like on one bond's cash flows.

Step 18. `bond_implied_curve` [DECK s.14] [ADAPTATION]

Purpose Turn one bond into a market credit curve, as slide 14 does.
Inputs Full price, bond terms, DF, R_b.
Formula Solve $\sum_i CF_i \text{DF}(t_i) e^{-ht_i} + R_b \int_0^T \text{DF}(s) h e^{-hs} ds = P$. Bond-implied par $s_b = (1 - R_b)h_b$.
Outputs h_b, s_b. Used by 30 and test C4.
Why Slide 14 gives the idea and no procedure. A flat hazard is the simplest reading. On 10/1/01 it gives 30.7% to 38.3% (price as clean or full), against about 31% on the chart.

Step 19. `hazard_multiplier` [ADAPTATION]

Purpose Compare a bond and an option at the same horizon.
Inputs Q, full price, R_b.
Formula Solve κ so that Q^κ prices the bond. $\text{PD}_{\text{bond}}(T) = 1 - Q(T)^\kappa$.
Outputs κ, $\text{PD}_{\text{bond}}(T)$. Used by 30.
Why One bond cannot give a curve. The model's shape carries the bond's level to the option's maturity. Chan et al. (2023, p.190) show maturity mismatch matters. $\kappa > 1$ means the bond is wider than options imply.

Step 20. `implied_recovery` [ADAPTATION]

Purpose Ask what recovery would close the gap.
Inputs Q, full price.
Formula Solve R^* so that step 15 equals the full price. Flat form: use hazard λ_p from step 22 in step 18's price, and solve for R^*.
Outputs R^*, flag if outside $[0, 0.8]$. Used by 30.
Why Slide 20 calls recovery "a big assumption". Bégin et al. (2023, pp.123, 125) extract recovery from put-implied intensity and CDS as $R = 1 - \text{CDS}/\lambda$. This is the same idea on a bond. They find under 1% of values negative (p.126, note 4). The bounds here are my judgement.

Step 21. `bond_implied_vol` [ADAPTATION]

Purpose Express the bond's spread in vol units.
Inputs $\sigma_{\text{cum}}(\cdot)$, $\bar{B}$, full price, R_b.
Formula Solve c so that $\sigma_{\text{cum}}(\cdot) \to c\,\sigma_{\text{cum}}(\cdot)$ prices the bond. Scan for all roots. Use the status codes of `implied_vol_bond`.
Outputs c, status. Used by 30.
Why CreditGrades (p.13) and Yu (PDF p.10) define this for CDS. `implied_vol_bond` assumes flat vol, so it needs a scale on the curve. A bond can have two roots near recovery.

Step 22. `option_implied_default` [DECK s.11] [ADAPTATION]

Purpose Extract default probability from puts in two ways.
Inputs Step 13, American mids, Δ_{BS}, DF.
Formula Model route: $\text{PD}_{\text{opt}}(T) = 1 - Q(T)$. Model-free route: take the lowest strike with $|\Delta| \le 0.15$ and mid ≥ 0.15. $U_p = P_A(K)/K$. Solve $U_p = \lambda_p(1 - e^{-(r+\lambda_p)T})/(r + \lambda_p)$. $\text{PD}_p = 1 - e^{-\lambda_p T}$.
Outputs $\text{PD}_{\text{opt}}(T)$, U_p, λ_p, PD_p. Used by 30.
Why The model-free route is Carr and Wu (2011, Eqs. 1, 4, pp.479–480). It assumes a default corridor, and Finkelstein's model has none (see Section 7.2). The gap between the two routes measures that.

Step 23. `stock_only_curve` [DECK s.12, 13, 14, 15] [ADAPTATION]

Purpose Compute the model the stock alone implies, with parameters held fixed.
Inputs $B_0 = \bar{B}$ and flat $\sigma_0 = \sigma_{\text{cum}}(1)$ fixed on a start date, S_t. Refit every 250 trading days. Variant: step 13a with rolling 3-year vol, which needs no options.
Formula Q from step 13 with today's S_t, then step 15.
Outputs Z_{stk}. Used by 24.
Why This is the deck's own test design: "Using the same model parameters for different dates" (s.12), $B = 6$, $\sigma = 42\%$ held for 15 months (s.13). The 250-day refit is my judgement.

Step 24. `three_gaps` [ADAPTATION]

Purpose Measure the three gaps.
Inputs $Z_{\text{mkt}}, Z_{\text{opt}}, Z_{\text{stk}}$. `log_gap`.
Formula $G_{bp} = \ln(Z_{\text{mkt}}/Z_{\text{opt}})$. $G_{ps} = \ln(Z_{\text{opt}}/Z_{\text{stk}})$. $G_{bs} = \ln(Z_{\text{mkt}}/Z_{\text{stk}}) = G_{bp} + G_{ps}$.
Outputs Three gaps per bond per day. Used by 25, 30, 33.
Why Slide 26 speaks of one residual. Three markets give two independent ones, and the pair tells which market moved. A positive gap means the first market is wider.

Step 24a. `des_gap` [ADAPTATION]

Purpose Compute the published bond-against-stock gap and rank it.
Inputs Step 13a in step 15, recovery $\alpha\bar{L} = 0.35$, market full price, lagged bond value.
Formula DES $= CS_D - CS_E$ in bp per bond. Each spread is the bond yield less the yield of a synthetic Treasury with the same cash flows. Average to the firm with lagged bond value as weight. Sort firms into quintiles each month. Accounting data lag at least 2 months.
Outputs DES, firm and bond level, quintile. Used by 30, 34.
Why This is Chen, Chen and Li (pp.2015–2016). It is $Z_{\text{mkt}} - Z_{\text{stk}}$ in levels, up to the choice of spread. Their typical quintile values are −139, +45, +103, +156 and +271 bp (p.2023). Note the middle firm has a gap of +103 bp, not zero.

Step 24b. `impvol_change` [ADAPTATION]

Purpose Measure the monthly rise in 1-year implied vol.
Inputs Month-end ATM call and put implied vol, interpolated to 365 days.
Formula $\Delta\text{ImpVOL}_t = \frac{1}{2}(\Delta\text{CVOL}_t + \Delta\text{PVOL}_t)$. Rank into deciles each month.
Outputs ΔImpVOL, decile. Used by 30.
Why This is Cao et al. (2023, pp.1375–1376, 1379). It uses options the bond has not yet priced. It is the time-series counterpart of a rising G_{ps}.

Step 25. `control_gaps` [ADAPTATION] [OWN FILES]

Purpose Take out liquidity and common moves.
Inputs Gaps, peers, sibling, illiquidity fields, S/D.
Formula (1) Subtract `peer_median_gap`. (2) Require `sibling_confirms`. (3) Own-files variant: each week regress $\ln c$ on $\ln(S/D)$ across names, and keep the residual. (4) Variant: regress the gap on bid-ask, log size and age. Check VIF first, since these fields correlate. (5) Variant from your basis note: regress on credit index and swap spread, fit a mean-reverting model to the residual, and use the simulated expected capture as the signal.
Outputs Adjusted gaps. Used by 30.
Why The model has no input for supply, demand or the basis (CreditGrades p.32). Most investment-grade spread is not credit risk (literature review, line 81). The own-files agent reads the cross-section's free intercept as absorbing an average bond premium. Nothing has tested that on bonds. Chen, Chen and Li find the bond premium survives illiquidity and jump controls (p.2031), so controls may not remove all of it. Variant (5) is from the scanned *3 Steps of CDS Bond Basis*, which has no stated author and is for a CDS basis.

Stage F. Risk

Step 26. `fk_greeks` [DECK s.20] [ADAPTATION]

Purpose Compute delta, gamma, vega and jump to default per instrument.
Inputs Steps 14, 15, 16. Position.
Formula δ, γ: bump S by ±1%, holding $\sigma_{\text{cum}}(\cdot)$, $\bar{B}$, λ. ν: scale $\sigma_{\text{cum}}(\cdot)$ by 1 ± 0.001. Jump J: bond $-(P - 100R_b)$, CDS seller $-100(1 - R)$, put Ke^{-rT} − Put, stock $-S$, call −Call.

Outputs (δ, γ, ν, J) per unit, and the put-only ratio $-\delta_P/\delta_p$. Used by 27, 28, 29, 31.

Why Slide 20 names the sensitivities and does not define them. Asset vol held fixed gives -26.9 on slide 21 against about -27.4. Equity vol held fixed gives -11.1. So the deck holds asset vol fixed. Leaving contract value out of the CDS jump fits slides 23–25 best. The put-only ratio follows Avino and Salvador (p.10), who hedge market value through the common state variable. Here $\partial/\partial V = \partial/\partial S$.

Step 27. `solve_hedge` [DECK s.21, 22, 23, 24]

Purpose Solve the hedge amounts.

Inputs Target greeks g, instrument greeks M.

Formula Solve $M\,n = -g$. Delta only: stock, or put alone. Delta and vega: put and stock. Delta and jump: put and stock. All three: put, stock, call. Guards [ADAPTATION]: reject if a put or call count is negative, or the stock dollar size exceeds 1.5 times the bond's value. Report the condition number, but do not filter on it.

Outputs $n = (n_p, n_s, n_c)$, status. Used by 31.

Why The guards are mine. The solve can fail. On GX with a 7.5 put and a 15 call it returns -12.7 puts and $-\$192$ of stock. The condition number cannot separate good from bad, since it is 759 here and 509 on the deck's own slide 24.

Step 28. `optimal_strike` [DECK s.25]

Purpose Use one put plus stock for all three exposures.

Inputs Steps 16, 26.

Formula Find K^* where the delta and vega solution also zeroes jump: $n(K)\,J_p(K) - n_s(K)S + g_J = 0$.

Outputs K^*, n_p, n_s. Used by 31.

Why Slide 25 gives $K^* = 39.2$ and no definition. This reading gives 39.5.

Step 29. `rehedge_due` [DECK s.20] [ADAPTATION]

Purpose Decide when to rebalance.

Inputs Held and target shares, days since last, S move, costs.

Formula Rebalance weekly, or if $|n_{\text{target}}/n_{\text{held}} - 1| > 10\%$. Compare with static [YU] and weekly [OWN FILES].

Outputs Flag. Used by 32.

Why Slide 20 says "not static" and sets no rule. Yu finds daily rebalancing does not help. Your master file refreshes weekly. The 10% is my judgement.

Stage G. Trades. Everything here is an addition to the deck

Step 30. `decide_trade` [ADAPTATION] [YU]

Purpose Turn gaps into one action.

Inputs Ranks, adjusted gaps, ΔImpVOL, κ, R^*, status flags, costs.

Formula Base: the monthly rank in step 24a. Top quintile is long bond, short stock. Bottom quintile is the reverse. Skip a long-bond entry if ΔImpVOL is in the top decile. Name-by-name variant: Table 3, with trigger g starting at $\ln 1.5 = 0.405$, in `peer_adjusted_signal`. Also `passes_cost_gate`, `sibling_confirms`, `vol_status_allows_trade`. Trade at `execution_day` with lag 1.

Outputs Trade type and side. Used by 31.

Why A rank has the published evidence, and it does not need a zero-mean gap. A fixed ratio test is biased at a fixed $\bar{L}$. In Chen, Chen and Li's middle quintile, $\ln(151.5/46.8) = 1.17$, well past 0.405 (p.2023). Yu's per-name fit of $\bar{L}$ removes that level. Trading the next day guards against price error entering both signal and return (paper_method Step 35). The ΔImpVOL skip is my addition.

Step 31. `size_trade` [ADAPTATION] [YU] [OWN FILES]

Purpose Size each leg.

Inputs Greeks, `cds_equivalent_face`, `rate_hedge_face`.

Formula Rank book: equal or lagged-value weight within a quintile, each leg to the same gross. Name-by-name: $F = A_{5y}/D_b$, then n from step 27, and $h = F \cdot$`rate_hedge_face`. Variant [OWN FILES]: size so that a 3 vol point convergence earns a set amount, capped.

Outputs F, n_s, n_p, n_c, h. Used by 32.

Why This keeps Yu's capital of 0.5 per unit of 5-year spread risk meaningful. Your master file sizes by target P&L over vega. The per-name cap there is \$10mm.

Step 32. `leg_pnl` [ADAPTATION]

Purpose Compute daily P&L per leg.
Inputs Sizes, marks, carry.
Formula Section 8. `daily_pnl` covers T1. T2 to T4 need signed stock and option legs, because `daily_pnl` ties the stock's sign to the bond's.
Outputs P&L per leg, account value by `next_account_value`. Used by 33, 34.

Step 33. `exit_rule` [YU] [ADAPTATION]

Purpose End the trade.
Inputs Gaps, days held, value, events, option expiry.
Formula Rank book: rebalance monthly. Test holding 1, 3 and 6 months. Otherwise `bond_exit_reason`, plus: option within 30 days of expiry (roll or close), and gap sign flip on G_{bp} for T2 and T4.
Outputs Exit reason. Used by 34.
Why Chen, Chen and Li find bond returns lose significance by month 3, and are near zero after a year (p.2035). The quintile gap is still above 200 bp two years on (p.2033). So a convergence exit will rarely fire, as in Yu's Table 3.

Step 34. `book_and_report` [YU] [OWN FILES]

Purpose Aggregate and report.
Inputs Closed trades, daily returns.
Formula `monthly_index`, `annual_sharpe`, `trade_summary`, `monthly_summary`, `attribute_pnl`, `break_even_cost` and `lag_sensitivity`. Report by trade type, direction, rating at entry. Also report capital from single-name default stress, as in your master file. If you test a grid of entry and exit rules, report the probability of backtest overfitting and the deflated Sharpe, as in your scanned basis note.
Outputs Section 6.

6 Master table of final outputs

Output	Meaning	Unit	Step	Frequency
$\bar{B}$, source flag	Barrier and how it was set	USD/share	9	per filing
$\sigma_{\text{cum}}(T)$, forward vols	Asset vol term structure	decimal	10	daily
$Q(T)$, $h(T)$	Survival curve and hazard	decimal	13	daily
Model par spread	CDS-equivalent on the model curve	bp	14	daily
$P_{\text{opt}}, Z_{\text{opt}}$	Bond value and spread on today's options	USD/100, bp	15	daily
$P_{\text{stk}}, Z_{\text{stk}}$	Same, on fixed parameters and today's S	USD/100, bp	23	daily
Put default part	$Ke^{-rT}(1-Q)$ by strike and maturity	USD/share	16	daily
Put diffusion part	The rest of the put	USD/share	16	daily
Default share	Default part over put price	decimal	16	daily
$\sigma^M(K,T)$	Model Black-Scholes vol, for skew comparison	decimal	16	daily
Z_{mkt}, full price	Market spread and price	bp, USD/100	17	daily
h_b, s_b	Flat bond hazard, bond-implied par spread	decimal, bp	18	daily
κ, $\text{PD}_{\text{bond}}(T)$	Hazard multiplier, bond default probability	decimal	19	daily
R^*	Recovery that closes the bond gap	decimal	20	daily
c, status	Bond-implied scale on asset vol	ratio	21	daily
$\text{PD}_{\text{opt}}(T)$, U_p, PD_p	Option default probability, model and model-free	decimal	22	daily
G_{bp}, G_{ps}, G_{bs}	The three log gaps, raw and adjusted	log ratio	24, 25	daily
DES, quintile	Published level gap, bond and firm	bp, rank	24a	monthly
ΔImpVOL, decile	Monthly rise in 1-year ATM vol	decimal, rank	24b	monthly
(δ, γ, ν, J)	Greeks and jump per instrument and position	per unit	26	daily
n, status, K^*	Hedge amounts, including put-only, and optimal strike	count	27, 28	at trade
Trade log	Type, side, dates, sizes, exit reason, P&L per leg	USD	30–33	per trade
Account value, return	Net of financing and costs	USD, decimal	32	daily
Monthly index	Compounded returns by trade type	decimal	34	monthly
Sharpe, t, drawdown	Annualised, net and gross, by group	number	34	per run
Attribution	Bond, stock, option, Treasury, financing, cost	USD	34	per run
Break-even cost	Cost multiple that takes the Sharpe to zero	multiple	34	per run
Empirical hedge slope	Stock beta of the rate-hedged bond, against model delta	number	34	per run

7 How to do each leg

7.1 Bonds

Which spread: Z-spread over the Treasury zero curve. It uses the bond's own cash flows and the whole curve, and the model can produce the same number by the same solver. Reasons and alternatives are in Part 3 of `02_bond_vol_method.md`. Compare Z only with Z. A bond's Z differs from a CDS spread for arithmetic reasons once the bond is away from par. In that note's example, the 5-year CDS is 167.6 bp while model Z runs from 155.9 bp at a 2% coupon to 177.0 bp at 9%.

Run the matched-Treasury spread as a check. Chen, Chen and Li use the bond yield less the yield of a synthetic Treasury with the same cash flows (pp.2015–2016). Their replication needs that form. Z-spread stays the base for name-by-name work, because the model and market use the same solver.

Keep the deck's bond-implied CDS as a second measure. Step 18 repeats slide 14. It lets you compare with the deck's charts and with any CDS you later obtain.

Turning the survival curve into a value for this bond. Use step 15. Each coupon and the principal count only if the firm survives to that date. Recovery pays at default. The κ, R^* and c of steps 19–21 restate the same price gap in hazard, recovery and vol units. On GX they are 0.908, 0.356 and 0.953 (Appendix A).

Recovery.

1. Use recovery of par, $R_b = 0.40$, for senior unsecured. A claim in bankruptcy is on par, not on market price.
2. Run the deck's values as sensitivities. Slide 14 shows the effect at GX's peak: R of 0.30 gives a model par near 37%, and R of 0.10 gives near 47.5%.
3. Use 0.35 in the published replication. Chen, Chen and Li set $\alpha = 0.7$ with $\bar{L} = 0.5$ (p.2015).
4. Keep R_b apart from $\bar{L}$. $\bar{L}$ is recovery on all liabilities and sets the barrier. R_b is recovery on this issue (Yu, PDF p.35).
5. Flag any R^* outside $[0, 0.8]$. That points to a wrong curve or a non-default premium, not to a real recovery.

Illiquidity and basis control. The model has no inputs for bond supply, demand or the basis (CreditGrades p.32). So:

1. Trade high yield first. Lardy's printed slide 25 shows correlation of spread changes rising with risk: 0.10 for AA, 0.32 for BBB, 0.50 for BB and 0.72 for B. Your literature review says most investment-grade spread is not credit risk (line 81).
2. Subtract the peer median each day, by rating and maturity bucket (step 25).
3. Require the sibling bond to agree (step 25).
4. Run the own-files cross-section as a variant. Nothing has tested it on bonds. The original plan named it: "analysis of residuals in price term p(mkt) − p(EVol) allows to include securities other than CDS" (`03_bond_and_vol.md`).
5. Use CDS as a check if it exists, through `cds_confirms`. Kim, Li and Zhang (2016) find the residual basis predicts bond returns (1.79% at 20 days). So a bond-only gap may be a basis trade, not an equity-credit one.
6. Expect the premium to sit where costs are highest. Chen, Chen and Li report −2.69% a year for large bonds against −6.16% for small ones (p.2037). So report results by bond size, and trust the large-bond number.

Callable bonds. High yield names often have only callable bonds. Part 1 of the note keeps one only if it trades 5 points or more under the next call price, valued to maturity. Measure how many names survive that screen before relying on it.

Reuse from `e2c_bond.py`. All of Part 1 screens, price cleaning, Z solver, duration, Treasury hedge, financing, costs and exits. **Do not reuse** `survival`, `risky_bond_price`, `bond_equity_delta` or `run_bond_backtest`, since they assume one flat vol. Add a test that `fk_bond_price` matches `risky_bond_price` at flat vol with the remedy off.

7.2 Options

Decomposing a put. Step 16 gives the split. Default pays the strike, since the stock goes to zero (slide 1). The rest is a knock-out put on V. The default share falls as the strike rises and grows with maturity:

Strike	Put, 2.59y	Default part	Diffusion part	Default share	Share, 1.59y	BS vol, 2.59y
5	1.485	1.231	0.253	0.829	0.700	91.6%
7.5	2.515	1.847	0.668	0.734	0.570	83.5%
10	3.735	2.463	1.272	0.659	0.482	78.4%
12.5	5.118	3.078	2.040	0.601	0.421	75.0%
15	6.637	3.694	2.943	0.557	0.377	72.4%

GX on 6/14/01. $S = 10.5$, $B = 6$, $r = 5\%$. Asset vol 46.5% to 2.59y and 49.8% to 1.59y. All from the deck's parameters.

This means a low-strike put is mostly a credit instrument, and an at-the-money long-dated put is still more than half credit on this name. The deck's slide 1 makes the same point at 0.386 and 0.522 of the put. Its two figures do not follow one formula (Section 9).

Extracting default probability. Use two routes, and report both.

1. **Model route** [DECK s.11]. Fit the surface and read $1 - Q(T)$. It uses every usable strike.
2. **Model-free route** [ADAPTATION]. Take one deep out-of-the-money American put and use $U_p = P/K$, from Carr and Wu (2011). Their filter is bid > 0, open interest > 0, more than 360 days, strike $\leq \$5$, $|\Delta| \leq 15\%$, and highest open interest (p.482). Chan et al. (2023, pp.195–196) drop the \$5 cap because under 1% of their puts met it. Bégin et al. (2023, p.126) use 6 to 18 months and keep $|\Delta| \leq 15\%$. Low prices are mostly tick size, so Carr and Wu also require a URC value of at least 3%, which is \$0.15 on a \$5 strike (p.483).

The two routes disagree by construction. Carr and Wu need a default corridor: the stock jumps from above B to below A. They state that a continuous barrier has no corridor (p.475). Finkelstein's barrier at $\lambda = 0$ is continuous. So under his model $P/(K\,\mathrm{DF})$ overstates default probability. It only reaches the true value as $K \to 0$:

K (GX, 2.59y)	0.1	1	2.5	5	7.5	10
$P/(K\,\mathrm{DF})$	0.280	0.284	0.299	0.338	0.382	0.425
BS delta of the put			−0.038	−0.092	−0.154	−0.220

True default probability in the model: 0.280.

A direct test settles which world you are in. Carr and Wu (p.482) say American put prices are linear in strike inside a corridor. Finkelstein's model says they are convex at every strike. Test $P(K)/K$ across the lowest listed strikes. Flat supports the corridor. Rising supports the barrier. The $K = 5$ put on GX passes Carr and Wu's delta filter and still overstates: $\mathrm{PD}_p = 0.315$ against the model's 0.280.

Which strikes and maturities.

- ATM vols at every listed maturity from 0.5 years, for the vol curve.
- At least 3 strikes below the money, in at least 2 maturities, for the barrier. The deck's strikes run to 0.48 of spot.
- Options past 2 years are thin. In Chan et al.'s 2012–2016 sample, 77% of such puts have under 6 months left, and 1.6% have over 2 years (Table 2, p.196). So the curve past the last option date is an assumption. Report how much of the bond's value depends on it.

Options play two roles, and they want different contracts.

- **To calibrate,** use long-dated options. The deck's GX table runs to 2.59 years. They carry the most default value.
- **To hedge,** short-dated options are what trades. Avino and Salvador use puts with a mean of 43 days, moneyness 0.946 and delta −0.30 (pp.15–16). Their ratio allows a short put against long debt, so the mismatch sits in the model, not in the signal (p.15, note 11).
- **Short-dated puts cost a lot to hold.** Their mean monthly put return is about −17% to −24% (p.16).

American against European. The deck does not address this. Three points decide it.

1. Under the model, an American put pays K at default, not at expiry. On GX this adds 0.021 to 0.125, which is 1.8% to 4.2% of the put. So treating an American price as European overstates default probability by a similar share.
2. A standard tree has no default, so it may remove too little of that premium.
3. The base case fits the European closed form to de-Americanised prices, with a default-leg correction $K(U - \mathrm{DF} \cdot \mathrm{PD})$. WP8 builds a grid pricer on V with an absorbing barrier and measures the remaining error.

Avino and Salvador convert to European with the method of Trolle and Schwartz (2009), and argue that early exercise matters little for short-dated puts (p.14, note 10; p.15). That argument does not reach long-dated puts near a barrier.

Dividends and borrow. Use the implied forward (step 5) for $\mu = r - p$. Use discrete dividends in the tree. Carr and Wu (p.479) assume dividends stop after default. A distressed name's forward may already show that.

Comparing option-implied and bond-implied default.

1. Common horizon: the longest usable option maturity T^*.
2. Option side: $\mathrm{PD}_{\mathrm{opt}}(T^*)$ and PD_p.
3. Bond side: $\mathrm{PD}_{\mathrm{bond}}(T^*)$ by κ, and $1 - e^{-h_b T^*}$ by flat hazard.
4. Report the log ratio at $R_b = 0.10$, 0.30, 0.40 and 0.50, since the bond side depends on R_b and the option side does not.
5. Expect noise. Bégin et al. (p.127) find put-implied default probabilities more volatile than CDS-implied ones.
6. Expect a bias. Chan et al. (Table 10, p.206) find put-implied hazards above CDS-implied among traded pairs (fitted means 0.0458 against 0.0175). Carr and Wu find the opposite sign on average, −1.56% in URC value, disappearing after mid-2007 (p.488). So estimate the level and trade changes against it.

7.3 Cash equity

Delta from the model. $\delta_P = \partial P / \partial S$ holding $\sigma_{\mathrm{cum}}(\cdot)$, $\bar{B}$, λ fixed, so V_0 moves one for one with S. A long bond shorts $F\delta_P$ shares. This is the deck's convention on my evidence: it reproduces slide 21 within 2%, and holding equity vol fixed misses by a factor of 2.4. Yu does not say which he holds. CreditGrades holds asset vol (p.36).

How the barrier changes delta as S falls. The share count rises all the way down. The dollar hedge peaks near $S \approx B$, then falls, because each share is worth less. Gamma stays negative for a long-credit position, so the hedge shorts more as the stock falls. It sells low and buys back high.

S	Par (bp)	Shares to short	Dollar hedge	Jump to default	Hedge / jump	Gamma
60	202	0.294	17.64	105.27	0.168	−0.011
40	408	0.673	26.93	96.22	0.280	−0.030
30	620	1.080	32.39	87.65	0.370	−0.054
20	1,032	1.826	36.51	73.52	0.497	−0.102
15	1,423	2.437	36.56	62.95	0.581	−0.146
10	2,164	3.332	33.32	48.68	0.684	−0.218
5	4,259	4.711	23.56	28.84	0.817	−0.342
2	10,082	5.824	11.65	13.06	0.892	−0.378

Slide 21 example: \$100 sold at 560 bp, $B = 15$, $\sigma = 37.5\%$, $R = 0.10$, $\lambda = 0$, $r = 5\%$. Jump here includes contract value.

This table is the main warning for the stock leg. A delta hedge does not hedge default. It covers 17% of the jump far from the barrier and 89% close to it. Puts must cover the rest. That is why slides 23–25 buy puts.

Rebalancing. Slide 20 says the hedge "is not static" and that costs count. Run static [YU], weekly [OWN FILES] and a 10% band [ADAPTATION]. Charge the stock half-spread on each trade. On a bond with a short hedge, every rebalance into a falling stock costs more, and the borrow may get harder.

Borrow cost. A short earns the cash rate less the fee, and pays dividends. Skip names flagged hard to borrow. Watch the implied borrow from step 5. It tends to rise as S falls, which is when the hedge needs the most shares. That last point is my expectation and not from a source. Test it. Chen, Chen and Li's stock premium is 16.89% a year in high-fee stocks and 2.52% in low-fee stocks (p.2037). So the short leg's edge sits largely where shorting costs most. Their paper does not report returns net of the fee.

The short-stock jump in default. Per slide 1 the stock goes to zero. The short gains $n_s S$. So the net position in default is $n_s S - F(P - 100R_b)$. Two points follow.

- At $\lambda = 0$ the model reaches zero continuously, so the jump is a scenario, not a model outcome. Treat it as an instant default from today's S.
- The stock may not reach zero at once. Carr and Wu allow a positive post-default level A (p.475). Run the stress at $S_{\mathrm{post}} = A$.

8 The trades

Source. Hedge structures come from the deck, slides 21–25. Every signal, entry, exit and size is an addition. T0 rests on later literature and has no hedge from the deck. The deck's hedges are for a CDS book. Using them on a bond is an adaptation. A long bond has the same sign of greeks as a sold CDS, and a par bond's delta equals the CDS delta (Part 4 of `02_bond_vol_method.md`).

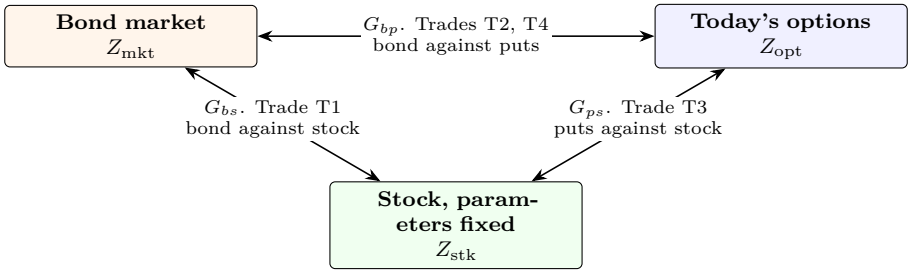


Figure 2: Three markets, two independent gaps: $G_{bs} = G_{bp} + G_{ps}$. If only one edge fires, that market pair has a disagreement the third does not share.

G_{bp}	G_{bs}	Reading	Action
$> +g$	$> +g$	Bond wide against both. The bond is the outlier	Long bond: T1, or T2 or T4 if puts are liquid
$< -g$	$< -g$	Bond tight against both	Short bond: T1 reversed, if the bond can be borrowed
≈ 0	$> +g$	Bond and options agree. Fixed σ_0 is stale	No trade. Update σ_0. This is Yu's "vol estimate too low" case
$> +g$	≈ 0	Options low against bond and stock	Buy vol: T3 long side, or T2
$< -g$	≈ 0	Options high against bond and stock	Sell vol: T3 short side, with a jump limit

Table 3: Decision table for step 30. g starts at 0.405. [ADAPTATION]

Evidence on direction. The strongest is Chen, Chen and Li (2026). In 1980–2022, the top DES quintile of bonds beats the bottom, and the bottom quintile of stocks beats the top. So a wide bond against the stock means long bond, short stock, which is the direction of T1 and of Yu.

- **Stocks.** Low minus high: 6.56% a year, $t = 3.79$ value-weighted and 3.51 equal-weighted (p.2007).
- **Bonds, full sample.** High minus low: 5.28% a year, $t = 5.64$ value-weighted, and 6.58%, $t = 6.02$ equal-weighted, in Table IV (p.2026). The paper's introduction prints 5.57% and 5.25% for the same sort (p.2007). I use the table.
- **Bonds, clean returns, 2002–2021.** 3.24% a year, $t = 2.69$, value-weighted (p.2020, note 9). This is the number to plan around. It is gross.
- **What the paper does not show.** I find no return net of cost and no combined stock-and-bond portfolio in its text. I did not read its Internet Appendix.
- **Two other reads.** Cao et al. (2023) find a 0.60% monthly lag in bonds after a rise in implied vol, and they find no such effect in stocks. Kapadia and Pu (2012, p.544) find convergence trades earned about 6% a year over 2001–2009. Carr and Wu (pp.492–494) find deviations predict both markets.

These are reasons to test, not expected results.

P&L, per leg, close to close. $s = +1$ for a long bond. n_s, n_p, n_c are signed. ρ_b is repo for a long and repo less fee for a short. ρ_s is cash less fee if $n_s < 0$, and cash if $n_s > 0$.

$$
\begin{aligned}
\text{Bond} &= sF\,(P_t - P_{t-1} + \text{coupon}_t) - sF\,P_{t-1}\rho_b\Delta t \\
\text{Stock} &= n_s(S_t - S_{t-1} + \text{div}_t) - n_s S_{t-1}\rho_s\Delta t \\
\text{Put} &= n_p(\text{Put}_t - \text{Put}_{t-1}) - n_p\text{Put}_{t-1}\,r_{\text{cash}}\Delta t, \quad \text{likewise Call} \\
\text{Treasury} &= -sh\,(T_t - T_{t-1} + \text{coupon}_t^T) + sh\,T_{t-1}\,\text{repo}\,\Delta t \\
\text{Costs} &= \text{half-spread on each leg, at entry, exit, every rebalance and every option roll}
\end{aligned}
$$

T0. Monthly rank book of bonds and stocks. Build first

Source: signal and sort [ADAPTATION], from Chen, Chen and Li (2026). No part of this is in the deck. It uses the deck's idea of comparing the formula with the market (slide 26).

Legs Long top-quintile bonds, short bottom-quintile bonds. Short top-quintile stocks, long bottom-quintile stocks. Treasury overlay on net duration.

Size Equal or lagged-value weight in each quintile. The paper reports each market as its own long-short portfolio. Running both together is my addition, so test the bond leg and the stock leg apart first.

Hedge Across names only. No name-level delta hedge in the base case. Rates by duration.

Carry Bond: coupon less repo on longs, coupon and fee on shorts. Stock: rebate less fee on shorts, financing on longs.

Signal Step 24a quintile, with `sibling_confirms` and the ΔImpVOL skip as variants.

Exit Rebalance monthly. Test 1, 3 and 6 month holds, and hold-off bands to cut turnover.

Goes wrong (1) Costs: Cao et al. (p.1394) use bond bid-ask costs of 18 bp for $1mm trades and 68 bp for $100k, and their signal fails at the higher number. (2) Short bonds may not be borrowable. Without them, the bond leg is a long-only tilt with credit beta. (3) The edge concentrates in small bonds and high-fee stocks. (4) A single default or buyout is not hedged by name. (5) The stock leg may be partly a value premium (p.2035).

T1. Bond against stock

Source: hedge [DECK s.21]. Signal form [YU]. Bond version [ADAPTATION]. Filter [ADAPTATION].

Legs Long bond, short stock, short Treasury. Reverse on the other side.

Size $F = A_{5y}/D_b$. $n_s = -sF\delta_P$. $h = F \cdot$ `rate_hedge_face`.

Hedge Stock for delta. Treasury for rates. No vega or jump hedge.

Carry Long side: coupon, less repo on P, plus rebate on the short, less dividends. GX before the stock: $9.50 - 4.24 = +5.26$ a year per $100.

Signal G_{bs} beyond g, with G_{bp} of the same sign. The filter rejects a stale σ_0. A name with no usable options runs on G_{bs} alone with the fallback vol, which is close to Yu.

Exit Gap crosses zero, 180 days, account below zero, or a forced event, by `bond_exit_reason`. Your plan document adds a stop at 3 times the target, and Kim's scanned deck suggests a 5 standard deviation stop. Neither was tested, so run them as variants [OWN FILES]. Test a 3-month hold beside 180 days.

Goes wrong (1) Default: the hedge covers only part of the jump, 25% to 93% on the GX bond. (2) A buyout lifts the stock and hurts the bond, Yu's "sure case of divergence". (3) Short gamma as S falls. (4) Bond costs: a typical Yu trade earned 1% to 3% on capital, and the planning cost is also 1% to 3%.

T2. Bond against put

Source: structure [DECK s.23]. Signal [ADAPTATION], on the idea of Carr and Wu and Chan et al.

Legs Long bond, long puts, stock to zero the delta, short Treasury.

Size Default-loss form, slide 23: solve $\begin{pmatrix} \delta_p & 1 \\ J_p & -S \end{pmatrix} \begin{pmatrix} n_p \\ n_s \end{pmatrix} = \begin{pmatrix} -F\delta_P \\ F(P - 100R_b) \end{pmatrix}$, with F in units of $100 of face. Market-value form, Avino and Salvador: $n_p = F\delta_P/|\delta_p|$, with no stock.

Compare Near the barrier the two forms agree. On GX, the market-value form takes 12.04 of the 7.5 puts ($30.3), and they cover 95% of the jump. Further away they differ. On slide 21 it takes 2.90 puts ($15.1) and covers 71%. In Avino and Salvador's mostly BBB and A sample the gap is wide: 94 puts costing $13,283 against 1,064 puts costing $122,605 per $10mm (p.93). So their market-value hedge is 89% cheaper, and it leaves most of the default loss open. Their default-loss hedge, in turn, raises hedging error on market value by well over 100% (p.76). Pick the form by the risk you want flat. For a trade that may hold to default, use slide 23. For a short convergence trade, use the market-value form.

Hedge Delta and jump are flat at entry. Vega stays open. On GX it is net short: $-62.25 + 13.63 \times 3.50 = -14.5$ per 1.00 scale of vol.

Carry Usually negative. GX per $100, $K = 7.5$, 2.59y: bond +5.26, put theta $13.63 \times (-0.506) = -6.90$, stock financing −0.19. Total about −1.83 a year, before gamma gains.

Signal $G_{bp} > g$ on Z_{mkt}, confirmed by $\text{PD}_{\text{bond}} > \text{PD}_{\text{opt}}$ at T^*. Variant: combine stock and put by the change in the vol skew (Avino and Salvador, pp.5, 35).

Exit G_{bp} crosses zero. 30 days before put expiry, roll or close. Also T1's exits.

Goes wrong (1) Term mismatch: the put expires first, so later defaults are unhedged. (2) Costs: Chan et al. report mean put bid-ask of 0.09 on a mean price of 0.44 (p.196), and their simple strategy goes from +18.3% to −8% a trade after costs (pp.205–206). (3) Only about one third of pairs converge (Chan, p.189). (4) A low-delta put can lose while credit is unchanged, if vol falls. (5) The old edge may be gone: practitioners viewed sell-CDS, buy-put as profitable, and Carr and Wu find the bias disappears after mid-2007 (p.488). (6) The hedging gain is modest. Model put ratios cut error by 19% against 14% for stock, and empirical ratios by about 25% for both (Avino and Salvador, p.29). In their lowest-rated group, stock does up to 10% better.

T3. Put against stock, with bond as the reference

Source: idea [DECK s.2, 9], the "flip side". Structure and signal [ADAPTATION].

Legs Low-strike put against ATM put, delta-hedged with stock. No bond position.

Reference B_{credit}: the barrier at which step 15 reprices the bond, at today's ATM-fitted vol. It implies a model skew $\sigma^M(K; B_{\text{credit}})$ (slide 9).

Size Choose vega target N. $n_{\text{low}} = N/\nu_{\text{low}}$. $n_{\text{ATM}} = -n_{\text{low}}\nu_{\text{low}}/\nu_{\text{ATM}}$. $n_s = -(n_{\text{low}}\delta_{\text{low}} + n_{\text{ATM}}\delta_{\text{ATM}})$.

Signal $\sigma_{\text{mkt}}(K) - \sigma^M(K; B_{\text{credit}})$ beyond 3 vol points, in either direction. 3 is my judgement. Slide 8's model-to-market gaps run up to 2.1 vol points.

Carry Option theta net of the ATM leg, plus stock financing.

Exit Vol gap inside 1 point, 30 days before expiry, or a jump-loss limit.

Goes wrong (1) A short low-strike put loses about Ke^{-rT} in default. Cap it. (2) Skew also carries demand and a vol risk premium, not only credit. Cremers et al. (PDF p.9) find individual skew very volatile. (3) In your 2007 screen, risk reversal passed the fit test on 229 names, fewer than stock (311) or ATM vol (308).

T4. All three together

Source: structure [DECK s.24, 25]. Applied to a bond and with a signal [ADAPTATION].

Legs Long bond, long put, short stock, long call (slide 24). Or long bond, long put at K^*, long stock (slide 25).

Size Solve $M\,n = -g$ with rows for delta, vega and jump, and columns for put, stock and call.

Hedge Delta, vega and jump are flat at entry. The residual is the spread gap itself, gamma, recovery and term mismatch.

Carry Coupon less financing, less option theta on two legs. Usually the most negative of the four.

Signal Same as T2. Use T4 when the guards in step 27 pass.

Exit Same as T2. Rebalance option legs only if the residual vega exceeds a limit.

Goes wrong (1) The solve can fail: GX with a 7.5 put and a 15 call fails the sign check. The deck's example works at $S/B = 2.67$, and GX sits at 1.75. (2) Four legs mean four costs. (3) Greeks are only flat at the starting S. (4) Slide 25 needs a listed strike near K^*.

Order of work. Build T0 first, because it has published targets and needs no options. T1 next, because the machinery exists. T2 is the deck's three-instrument idea in its simplest form, so build it third. T4 only if T2's residuals warrant it. T3 last, because it needs a clean skew. The variance swap as a vega hedge appears in JPMorgan's 2007 CEV note, in Berd and Voronov (2007) and in your 2007 screen. It is not in the deck, and I leave it out.

9 Acceptance tests

Every value below comes from `check_deck_numbers.py` unless marked "to build". Use $r = 5\%$ and $p = 0$ unless stated. The deck does not print r for these slides. A value of 5% fits slides 8, 12 and 21.

ID	Test and inputs	Expected	Tolerance
A. Limits and identities			
A1	$Q(0)$ from Eq. (1), remedy off. $\lambda = 0$. Then $S/B = 1$ and $S/B = 2$ at $\lambda = 0.3$ [DECK s.19]. $Q(0) = 1$ only at $\lambda = 0$	1. 0.9867. 0.999867	1e-4
A2	Q never rises with T	True	exact

continues

ID	Test and inputs	Expected	Tolerance
A3	`fk_survival` against `e2c_bond.survival`, flat vol, no remedy	Same	1e-12
A4	Eq. (1) equals slide 11 averaged over B, no cut-off. $S = 15, \bar{B} = 15, \sigma = .25, \lambda = .3, T = 1$	0.934494 both. With cut-off, step 13a at zero drift, 0.936004	1e-5
A5	Knock-out part: closed form against direct integration. $K = 7.5, T = 1.59, \sigma = .498, B = 6$	0.90310541	1e-7
A6	$B \to 0$: model put against Black-Scholes. $S = 10.5, K = 10, T = 1, \sigma = .5$	1.5191	1e-3
A7	$K \to 0$: $P/(K\text{DF})$ reaches PD. GX, 2.59y	0.2801	1e-3 at $K = 0.001$
A8	$K \to 0$ call: Se^{-pT}. $S = 10.5, T = 2, p = .02$	10.0883	1e-4
A9	Slide 6 chart at $S = 2$ and $S = 12$	165.0%, 73.3%	0.1 pt
A10	Remedy: $Q(0) = 1$, Q continuous at 0.5, flat hazard below	True	1e-12
A11	Step 13a at zero drift, $T = 1$, $\bar{B} = 15$. $S = 30$, then $S = 5$	0.990967. 0.640711	1e-5
A12	Step 11 variant as a share of σ_S. $S/D = 4, 1, .5, .25$	0.800, 0.600, 0.467, 0.360	1e-3
B. Options against the deck			
B1	Slide 8, ten model vols. $S = 10.5, B = 6$, vols 49.8%, 46.5%	Deck's column	0.15 vol pts each
B2	Fit market vols of slide 8, one B, two σ	$B = 6.01$, 0.500, 0.469, rmse 0.81	$B \pm 0.25$, $\sigma \pm 0.01$
B3	Forward vol, 1.59y to 2.59y	40.7%	0.1 pt
B4	Slide 1 from two vols only, 72%, 77%. $r = 3\%$, then $r = 5\%$	$B = 6.95$, PD .1496, par 544 bp. $B = 6.80$, PD .1440, par 519 bp. Deck: PD .136, par 530 bp	± 0.05 on B, ± 0.001 on PD, ± 2 bp
B5	Slide 1 from ATM 72% and PD .136: predicted OTM vol	76.6% at $r = 3\%$, 76.7% at $r = 5\%$. Deck 77%	0.1 pt against mine, 0.5 against deck
B6	Slide 7. $K/F \geq 0.5$, $r = 0$	$B = 10$: .99, .82, .74, .70. $B = 5$: .94, .82, .77, .74. $B = 1$: .85, .82, .81, .80. Chart ± 0.05	0.01 against mine
C. Credit against the deck			
C1	Slide 12 par curve at ten tenors. Bootstrapped vol, $R = .3$, $\lambda = 0$	0.93, 4.84, 7.59, 8.29, 8.54, 8.67, 8.75, 8.71, 8.52, 8.27 (%)	20 bp against chart readings
C2	Slide 12 bond, clean, flat 42%, $R = .3$	83.4, 59.9, 44.5. Deck 82, 60, 43	1.5 points
C3	Slide 14 peak, $S = 1.8$, $\sigma = 42\%$, to 1/3/06	37.7% at $R = .3$, 48.5% at $R = .1$. Chart $\sim$37, $\sim$47.5	1.5 points
C4	Slide 14–15, flat hazard fit at 10/1/01, $R = .3$	Par 30.7% to 38.3%. PV -49.6 to -54.5. Chart $\sim$31%, ~ -53	Chart inside the bracket
C5	Slide 19 left, printed inputs	rms 50 bp to chart. At $\sigma = 42\%$: 17 bp	$\leq$60 bp
C6	Slide 19 right, $S/B = .8$, $\sigma = 17.8\%$	rms 29 bp to chart. Printed 22.2%: 159 bp	$\leq$35 bp
C7	CreditGrades Table C.1 by `e2c_bond.check_table_c1` with $\times 360/365$. It keeps the mass at $T = 0$, so run it with the remedy off	156 of 156. Largest gap 0.494 bp. Without the scaling 11.59 bp	0.5 bp
C8	Five rows of C.1 by the deck's Eq. (1), atom kept, then remedy on	0.48 bp, then 0.99 bp	1 bp
D. Hedges against the deck			
D1	Slide 21 stock dollars at $\lambda = 0$, asset vol held. Then equity vol held	-26.9. -11.1. Deck ~ -27.4	Within 5% of the deck. Equity vol must fail
D2	Slides 21–25. $r = 5\%$, $p = 2\%$, $\lambda = .3$ on CDS, 0 on options, CDS jump 90, deck's option prices	Par 424. -26.3. 32.2, 29.7. 25.9, 18.8. 17.1, -32.5, 14.5. $K^* = 39.5$. 48.2, 43.7	8% against the deck's 422, -27.4, 32.86, 31.85, 26.4, 20.2, 17.54, -33.27, 14.68, 39.2, 49.13, 46.41
D3	Signs on slides 22, 23, 24, 25	$(+,+)$, $(+,+)$, $(+,-,+)$, $(+,+)$	exact
D4	After a solve, net delta, vega, jump	0	1e-8
D5	Delta path table in Section 7.3	Shares rise. Dollars peak at 36.6. Cover 0.168 to 0.892	1%
D6	Guard: GX 3-leg solve with 7.5 put, 15 call	Fails the sign check. -12.66 puts	Must be rejected
D7	Put-only hedge. Slide 21, then GX $K = 7.5$, 2.59y	2.90 puts, \$15.1, 71% of jump. 12.04 puts, \$30.3, 95% of jump	1%
E. Worked example, Appendix A			
E1	GX 6/14/01 spreads. Identity $G_{bs} = G_{bp} + G_{ps}$	724, 787, 736 bp. -0.084, $+0.066$, -0.017	1 bp. Identity 1e-10
E2	κ, R^*, c, h_b	0.908, 0.356, 0.953, 0.1077	0.002
E3	DES form on GX. Step 13a, $D = 14.2$, $\sigma_S = 82.2\%$, $R = .35$. Payout .03, then .05	-149 bp, ratio 0.919. -213 bp, ratio 0.888	2 bp
F. Pipeline tests, to build			

continues

ID	Test and inputs	Expected	Tolerance
F1	Quotes generated from known B, σ_{cum}, noise ± 1 vol pt, then calibrated	Recover B and PD	$B \pm 15\%$, PD ± 0.02
F2	Market bond price set to model price	No signal, zero trades. `check_no_signal_without_gap`	exact
F3	Known gap closes, nothing else moves. `check_known_gap_profit`, `check_legs_add_up`	P&L equals price change less costs, financing	1 cent
F4	Run to t, then $t + 100$. `check_no_lookahead`	Same results to t	exact
F5	Entry lag 0, 1, 2, 5. `lag_sensitivity`	If profit vanishes at lag 1, it was bounce	report
F6	Each bond on another issuer's stock and options. `placebo_panel`	No signal value	report
F7	American grid pricer at $r = 0$, no dividends	Equals the European price	1e-3
F8	Bond delta from data. `empirical_hedge_slope`	Compare with model delta	report
F9	On CDS if available: Lardy's printed slides 19, 25, 26	50% of relative basis within $\pm 20\%$. Change correlation rises with risk. Pairwise noise correlation near 0.15	Shape only
F10	Replicate Chen, Chen and Li on your sample	Quintile gaps -139 to $+271$ bp. Price ratio 0.94 to 1.16. Bonds up and stocks down in the top quintile. Bond premium near 3.24% a year after 2002	Signs must match. Sizes as shape, if samples differ
F11	Slope of credit change on put return times the model ratio	Near 1. Avino and Salvador: 1.02, t against one of 0.24, on CDS (p.23)	report
F12	Top minus bottom ΔImpVOL decile of bonds, next month	Negative. Cao et al.: -0.60%, $\lvert t \rvert = 3.62$	Sign must match

Deck numbers that do not reproduce, with the likely reason

The reasons are my reading. The deck does not confirm any of them.

1. **Puts 4.92 and 2.72.** At the printed 3% they come out 5.24 and 2.93. Annual or continuous compounding makes no real difference. A single rate of 4.78% gives 4.91 and 2.73. The rest of the deck fits a rate near 5%. So the printed IR may not be the rate used. The discount in the next line does use 3%.
2. **1.90.** The slide's own formula gives 1.92. It is 1.2% off, so this may be rounding.
3. **1.42.** The same formula gives 1.35. Any $K \times \text{PD} \times \text{DF}$ rule makes 1.42/1.90 equal 10.5/15, and it is 0.747 against 0.700. The undiscounted $10.5 \times 0.136 = 1.43$ is the nearest match.
4. **13.6% does reproduce, but only with annual premiums.** A flat hazard gives 14.05%. Quarterly gives 13.93%. Annual, default at period end, gives 13.58%.
5. **Slide 7.** Low-strike levels sit below mine by 0.1 to 0.2. They match if the strike axis is read one 0.05 step lower. The partly hidden caption does not describe the chart.
6. **Slide 10.** The asset vol line stays near 75% to 80%. With $B = 6.9$ and the charted ATM vol and stock price, I get about 54% by day 240. A B of 2 to 3 keeps asset vol flattest. So the slide's stability claim is not confirmed. It matters because slides 12–15 hold σ fixed for 15 months.
7. **Slide 19 right.** 22.2% is 40%/1.8, which treats 0.8 as B/S. The chart itself fits $S/B = 0.8$ with $\sigma = 17.8\%$.
8. **Slide 21's 422 bp** needs λ near 0.28. At $\lambda = 0$ it is 408 bp. The deck does not state λ there.
9. **Slide 12's third bond.** 46.3 full and 42.7 clean with the bootstrapped vol, against 43. No single convention hits all three dates. The slide does not say clean or full.

10 Build plan

First package: WP1. It costs little and everything depends on it. It turns the check script into a tested library. **First empirical run: WP6a.** It needs only bonds, stocks and accounts, and it has published targets, so it should come before any option data. After it, WP2, WP3, WP4 and WP8 are independent and can run in parallel. Sizes are rough estimates of focused work.

WP	Produces	Needs	Size	Done when
WP1	`fk_model.py`, tests. Steps 12, 13, 13a, 16. Black-Scholes helpers	Nothing	1–2 days	A1–A12, B1–B6 pass. No market data used

continues

WP	Produces	Needs	Size	Done when
WP2	`fk_value.py`. Steps 14, 15, 18–21, 23, 24	WP1, `e2c_bond`	1–2 days	C1–C8, E1–E3 pass. `fk_bond_price` equals `risky_bond_price` at flat vol with remedy off
WP3	`fk_risk.py`. Steps 26–29	WP1, WP2	1–2 days	D1–D7 pass
WP6a	`fk_des.py`. Steps 1, 2, 7, 11, 17, 24a. T0 returns. Replication on your sample, no options	WP2	3–5 days	F10 passes on signs. A note gives gross returns by bond size and fee, and net at measured costs. If signs fail, stop and debug before any option work
WP4	`fk_options.py`. Steps 3–6, on synthetic quotes first	WP1	2–3 days	Synthetic American quotes round-trip. Bad quotes rejected. Forward recovered within 1 bp of carry
WP5	`fk_calibrate.py`. Steps 7–11, 22	WP1, WP4	2 days	F1 passes. U_p matches the Carr-Wu formula on a hand case
WP6	`fk_gaps.py`. Steps 25, 24b, wired to steps 18–24	WP5, WP6a	2 days	Gaps run on a synthetic panel. F2 passes
WP7	`fk_trades.py`, `fk_backtest.py`. Steps 30–34. T1, then T2	WP3, WP6	3–5 days	F3–F6, F11, F12 pass for T1 and T2. Leg P&L sums to the account every day
WP8	`fk_american.py`. Grid pricer on V with absorbing barrier	WP1	2 days	F7 passes. A note states the de-American error on GX inputs, and whether the base case stands
WP9	First option run on 20–50 high yield names, with reports	WP4–WP7	2–4 days	A table of gaps, hedge slopes (F8), P/K linearity test, and asset vol stability. Slide 10's claim settled for this sample. A test of whether $Z_{\text{mkt}} - Z_{\text{opt}}$ adds to $Z_{\text{mkt}} - Z_{\text{stk}}$
WP10	Full backtest, T3 and T4 if earned, written report	WP8, WP9	1–2 weeks	Net Sharpe, t, drawdown, break-even cost, by trade type and rating. Costs at 1, 2 and 3 times measured. Any rule grid reported with overfitting probability and deflated Sharpe

Stop rules. Stop and report if WP6a's net bond return at measured costs is zero or negative at the large-bond end. Do the same if WP9's empirical hedge slope is far from model delta. A fixed trigger is not a good stop test, since its firing rate depends on the parameters. On GX the log gap is 0.02 to 0.08 in size on the deck's values, and 0.19 to 0.26 on the published settings.

11 Open questions and risks

Open questions, ranked, with my recommendation

1. **Signal: a monthly rank, or a fixed trigger per name?** This decides the whole trading layer. **Use the rank as the base, and keep the trigger as a variant.** The rank has the published result. A fixed ratio test is biased at a fixed $\bar{L}$, because the typical gap is not zero: Chen, Chen and Li's middle quintile has a log ratio of 1.17. Yu avoids that by fitting $\bar{L}$ to each name's first ten days, which adds a fitted number and turns the signal into a change since the fit. The gap's level also moves with parameters. On GX it is -12 bp on the deck's values, and -149 to -213 bp on the published ones. A rank should be less exposed to that, though I have not tested it.

2. **Barrier: from each name's skew, or $\bar{L}D$?** Every number depends on it, and the deck allows both. **Use $\bar{L}D$ with $\bar{L} = 0.5$ by default.** Chen, Chen and Li get their result with exactly that (p.2015). Use the skew fit only if it improves rmse by 0.5 vol points or more and stays in $[0.25D, D]$. On GX the fit gives 6.01 and $\bar{L}D$ gives 7.1. The rmse differs by about 0.16 vol points and the 2.59-year default probability by about 0.012. So the skew adds little where it can be read, and it will be noisier where quotes are thin. On GX this rule picks $B = 7.1$, not the deck's 6. Tests that reproduce the deck pass B in directly. Confirm in WP9.

3. **Daily refit, or hold parameters?** The deck does both: it fits on 6/14/01 and holds through 10/1/01. **Compute both.** Z_{opt} uses the refit, Z_{stk} uses the hold. On name-by-name trades, require both to agree. On the rank book, start with the Z_{stk} form and test whether the Z_{opt} form adds to it. The literature review (gap 4) says nobody has tested implied vol change, CDS momentum and the debt-equity spread together.

4. **λ in credit and in options?** The deck fits 0 on GX and about 0.28 on the hedge example. It prints no option formula with λ. **Use 0.3 on the credit curve with the slide 19 remedy, and 0 on options.** Slides 21–25 reproduce best this way. Run $\lambda = 0$ on credit as a sensitivity. It is an inconsistent pair of assumptions, and at short tenors it matters: 468 bp against 566 bp at 2 years on slide 19's left panel.

5. **Recovery? Use 0.40 for senior unsecured, 0.35 in the replication, report R^* daily, and run 0.10 to 0.50.** Bond-specific recovery needs a source I do not have. Bégin et al. (p.123) call a fixed recovery popular but inadequate.

6. **Which names? High yield first, BB and below, non-financial, not captive finance.** Lardy's correlation evidence and the literature review both point there. The own files show GM and Ford have no solution with this formula.

7. **American correction? Start with the tree plus default-leg correction, then let WP8 decide.** The effect is 1.8% to 4.2% of the put on GX.

8. **What about your cross-section? Run it as a variant of the rank.** Your weekly regression on $\ln(S/D)$ is your strongest result, at gross Sharpe 2.51. That falls to 0.62 at 5% round-trip costs at real turnover, with no hold-out. Chen, Chen and Li find a linear mix of model inputs does not reproduce DES (p.2031). So rank the model gap first, then test whether the regression residual adds to it.

Risks

Where the method is likely to fail.

1. **The published result may not transfer.** DES is a stock-only signal on 1980–2022 data, gross. Its bond premium on clean data after 2002 is about 60% of the full-sample figure. Nothing in these sources tests an option-calibrated model on bonds. Chen, Chen and Li cite one out-of-sample study (Deng, Hou and Shi, 2025), which I have not read.
2. **Thin evidence.** The deck tests two names. Both lost most of their equity value in the sample (slides 10, 13). A falling stock drives the model spread up in any structural model. So tracking on those paths does not show much about a name with a stable stock.
3. **Asset vol stability is unconfirmed.** Slide 10 does not reproduce at the end of its sample with its own B.
4. **Near the barrier.** Slide 18 excludes "super-depressed" stocks. At $S/\bar{B} = 0.33$, $T = 1$, Eq. (1) gives 0.601 and step 13a gives 0.641.
5. **Term mismatch.** Option data stop early, and bonds run longer. The long end of Q rests on a flat-forward assumption. On GX, flat 46.5% against the bootstrapped curve moves the 5-year par by 74 bp.
6. **Debt.** D is stale between filings, and it can hide off-balance-sheet debt (CreditGrades WorldCom case, p.35).
7. **Captive finance, financials.** No solution for GM, Ford and others [OWN FILES]. Yu excludes financials and utilities (PDF p.16).
8. **Events.** Buyouts lift the stock and hurt the bond (Yu, PDF p.12). Calls change the cash flows.
9. **Bond costs may exceed the edge.** See Part 9 of `02_bond_vol_method.md`. In the rerun of your workbooks, a one-week lag cuts Sharpe from 2.51 to 1.47.
10. **Model choice.** Ju et al. (2015) find an extended Geske-Johnson model beats CreditGrades for this trade on 369 obligors, 2004–2008. Finkelstein's model shares CreditGrades' constant barrier.

The deck's date and context. It is a client presentation from June 2002, by a firm that states it makes markets in the fixed income securities it discusses (PDF p.28). Its examples come from two names whose stocks collapsed in 2000–2001. It reports no costs. Its claim that basis risk and costs are "manageable" (slide 26) has no supporting number in the deck.

Market changes after 2002 that these sources support.

- **CDS contracts changed.** The market moved toward central clearing, upfront payments and fixed coupons (Carr and Wu, p.481, note 3, and Chan et al., p.194). A market par curve now needs conversion, if CDS is used as a check.
- **CDS shrank.** Notional peaked at USD 62 trillion at end 2007 and fell to USD 25 trillion by August 2012 (Avino and Lazar, 2020, p.75).
- **Option markets changed.** A penny pilot began (Carr and Wu, p.483, note 4). Deep out-of-the-money puts had two to four ticks of spread before it. A listed credit event binary option launched in 2007 and failed (p.476).
- **The classic trade decayed.** Standard capital structure arbitrage shows negative Sharpe ratios both before and during the 2007–2008 crisis (Avino and Lazar, p.83). Hedging with equity was effective only before the crisis (p.77). Yu's own Sharpes on 2001–2004 data were 0.15 to 0.89. The rank form looks better on one count: Chen, Chen and Li's bond premium stays significant after 2002, at a lower level, before costs. The samples and return definitions differ, so this is not a clean comparison.

- **Bond liquidity.** Dealers cut back liquidity as losses and funding pressure rise, and arbitrage unwinds can widen mispricing (Choi, Shachar and Shin, 2019, abstract). Bond price errors can create false premia (Dickerson et al. 2026, through `02_bond_vol_method.md`).
- **Limits to arbitrage.** Discrepancies are common: 41% of five-day moves (Kapadia and Pu, 2012, p.544). The authors link integration to liquidity and idiosyncratic risk.

A Worked example on the deck's numbers: GX on 14 June 2001

This shows the arithmetic only. It is not evidence the trade works. Inputs from the deck: $S = 10.5$, $B = 6$, $R = 0.30$, bond 9.5% of 11/15/2009 at 84, vols 49.8% and 46.5%, fixed $\sigma_0 = 42\%$. Assumed: $r = 5\%$, $p = 0$, $\lambda = 0$, 84 is a clean price. Full price is 84.77. Regenerate with section K of the check script.

1. **Three spreads.** $Z_{\text{mkt}} = 724$ bp. $Z_{\text{opt}} = 787$ bp, on the options curve (price 81.79). $Z_{\text{stk}} = 736$ bp, on fixed 42% (price 84.16).

2. **Three gaps.** $G_{bp} = -0.084$. $G_{ps} = +0.066$. $G_{bs} = -0.017$. The identity holds. All sit far inside 0.405. **So no trade on this date.** That fits the deck's message that the model tracks this name.

3. **Same gap in other units.** $\kappa = 0.908$. $R^* = 0.356$. $c = 0.953$, so bond-implied asset vol to 2.59y is 44.3%, against 46.5%.

4. **Default probability to 2.59y.** Options 0.280. Bond by κ 0.258. Bond by flat hazard 0.244. Carr-Wu on the $K = 5$ put 0.315.

5. **Bond risk per $100 face.** Delta 2.697 shares, or $28.32. Vega −62.25 per 1.00 scale of vol. Jump −51.79.

6. **Hedges with 2.59y puts.**

K	Put	Delta	Vega	Jump	Theta/yr	Delta + jump: puts, stock $	Delta + vega: puts, stock $
5	1.485	−0.149	2.60	2.908	−0.409	17.41, −1.2	23.96, 9.1
7.5	2.515	−0.224	3.50	4.073	−0.506	13.63, 3.7	17.78, 13.5
10	3.735	−0.297	4.14	5.049	−0.537	12.16, 9.6	15.05, 18.6

7. **Carry.** Bond 9.50 less financing 4.24 is +5.26 a year. The delta-and-jump hedge at $K = 7.5$ costs 6.90 of theta and 0.19 of stock financing. So the package runs at about −1.83 a year, before any gamma gains. A hedged trade must earn that back from the gap closing.

8. **Three legs fail here.** Put 7.5, stock and call 15 gives −12.66 puts, −$191.6 of stock and +$88.3 of calls. The guard rejects it.

9. **Puts alone, matched on delta.** 18.15 of the 5 puts ($26.9) cover 102% of the jump. 12.04 of the 7.5 puts ($30.3) cover 95%. 9.08 of the 10 puts ($33.9) cover 89%. Near the barrier, a delta match is nearly a jump match.

10. **The same day on published settings.** Use step 13a with $D = 14.2$, $\bar{L} = 0.5$, $\lambda = 0.3$, recovery 0.35, and ATM 82.2% standing in for 3-year vol. Asset vol is 48.9%. Model Z is 873 bp at payout 0.03 and 936 bp at payout 0.05, against the deck-parameter 787 bp. So DES is −149 to −213 bp, near the paper's lowest quintile value of −139 bp. The model price is 0.92 to 0.89 of market, where that quintile averages 0.94. On these settings GX's bond looks rich against its stock. On the deck's settings it does not. That parameter sensitivity is why the base signal is a rank.

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- Own files: `notes/own_models/02_implemented_method.md` and `03_bond_and_vol.md`, which summarise your workbooks and `CapitalStructureArb-1.doc`. I did not read the raw workbooks.
- Paper method: `notes/paper_method/02_bond_vol_method.md`, `e2c_bond.py`.
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Scanned files, read as page images. OCR on these is unusable.

- *3 Steps of CDS Bond Basis* (15 pages, no stated author, handwritten notes on it). Residual basis from index and swap betas, a fitted model, simulated expected capture as the signal, and overfitting checks across entry and exit grids. Used for step 25 variant (5) and step 34. It also has order-flow slides, which I do not use.
- *Automated Capital Structure Arbitrage* (J. Kim, 15 pages). CreditGrades against CDS, ranked across 1,000 names and 100 input series, with cointegration tests and a 5 standard deviation stop. Used for T1's stop variant. Its Johansen table shows rank 2 of 2 series, which means both series are stationary, not that they are cointegrated. Treat that claim with care, on 94 observations.
- *Empirical Models of Single- and Multi-Name Debt-Equity Relationship* (Berd and Voronov, 2007, 19 pages). Mostly market-level GARCH and correlation models. Its slide 5 argues for structural models with parameters set by the market, which fits Finkelstein. No number used.
- *Credit Grades Model Description* (one slide). The same content as CreditGrades. No number used.

Other files, no number used. *A Framework for Credit-Equity Investing* (Goulden, Allen and Einchcomb, JPMorgan, 19 November 2007; text search only) uses a CEV model, a z-score signal, and stock and variance swap hedges. It reports a 64% success rate and an information ratio of 1.96 in its own back-test (p.3). `Creditgrades_model.xlsm`: a one-name CreditGrades sheet, opened read-only with no macros. It uses ATM implied vol as its input vol.